

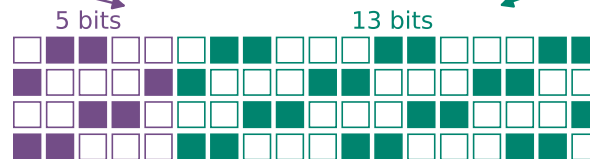
Amino-acid sequence



20 canonical amino acids

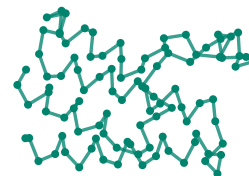
Amino-acid code → 5 bits

18 bits / residue
Compact shared interface

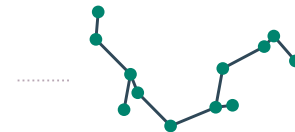


One row = one residue-aligned patch

Protein backbone geometry



Retained C α trace · 100 residues



N-C α -C, O

Frozen structure encoder + 13-sign LFQ

8,192 structural codes
Full LFQ vocabulary retained

One shared bitstream denoiser

One projection · one transformer · one bitwise head

Condition by clamping bits; denoise the remaining bits

Forward folding

Observed sequence

Generate structure

Inverse folding

Generate sequence

Observed structure

Joint generation

Generate sequence

Generate structure

One checkpoint · three tasks · no task-specific fine-tuning

Structural bits → frozen decoder → N, C α , C, O coordinates